

Chapter 13

Chronostratigraphy: understanding rocks and time

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13.1 Introduction

Chronostratigraphy is the branch of stratigraphy that deals with the relationship between rocks and time. It is key when considering and interpreting which time intervals rocks represent and for assessing uncertainties associated with correlating strata. A formal chronostratigraphical classification scheme is critical for standardization and communication. Ultimately, chronostratigraphy has led to the establishment of a globally applicable standard geological timescale (see also Chapter 15). Chronostratigraphy and the geological timescale are fundamental to the Earth (and planetary) sciences; they provide a framework within which all stratigraphical records can be placed and their construction has, and does, require the application and integration of a wide range of stratigraphical methods. Arguably, chronostratigraphy, and, more specifically, the development of the geological timescale, is one of the greatest achievements and largest collaborative projects of the geoscience community, representing the integration of 200 years of worldwide study, building global correlation frameworks that piece together fragments of evidence from the rock record from every continent. This was initially mostly achieved through the application of biostratigraphy for correlation (Chapter 4) and radio-isotope dating (Chapter 14) for numerical age calibration. But, increasingly, the timescale integrates other stratigraphical techniques that have become better developed, especially magnetostratigraphy (Chapter 5), chemostratigraphy (Chapter 6) and cyclostratigraphy (Chapter 9), in addition to more sophisticated integration and correlation techniques (e.g. Section 17.2; Chapter 15). Furthermore, advancements in our understanding of where there might be hiatuses and changes in sedimentation rate in the geological record (chapters 8 and 12) have helped the establishment of a robust chronostratigraphical framework. The resulting geological timescale and chronostratigraphical chart (Section 13.1.1, Section 15.2 and see the back inside cover) provide a temporal framework for the history of our planet (e.g. Benton and Harper

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2020), and the means to quantify rates of change in physical, chemical and biological processes (Section 15.2.1).

In essence, chronostratigraphical classification is relatively straightforward: there are rocks that accumulated over time, we need a global framework within which to organize this record, and we achieve this through the development of standards and classifications that allow us to communicate and apply this rock–time–age information. This chapter provides a basic description of how chronostratigraphy works and how we go about developing these international standards, which are the result of extensive, multidisciplinary and consensual global efforts.

13.1.1 Geological timescales and the International Chronostratigraphic Chart

Currently there are two main international geological timescales/chronostratigraphical charts: the International Commission on Stratigraphy's (ICS) International Chronostratigraphic Chart (http://www.stratigraphy.org: Cohen *et al.* 2013; Harper 2019) and *The Geologic Time Scale* (GTS2020: Gradstein *et al.* 2020 and earlier iterations). The ICS Chart shows the current state of chronostratigraphical classification and standards (Global Stratotype Section and Points (GSSPs): sections 13.4 and 13.5), and places emphasis on these GSSPs. GTS2020 is a comprehensive summary and integration of the latest timescale science including the work of ICS on the GSSPs (sections 13.3, 13.4 and 13.5.1).

Fundamental to any timescale is the way that all the data are scaled and integrated on a chronometric scale; however, depending on the methods and weightings applied, the final solution will vary in the detail. The different chronometric scaling methods that were chosen for each geological period in GTS 2020 are outlined in Gradstein *et al.* (2020), but there are other solutions: for instance, the Malinverno *et al.* (2020) polarity timescale (Chapter 15).

Capitalization

At any level in the chronostratigraphical classification hierarchy, an initial capital letter is used for each and every formal component of the name: for example, Jurassic System, Lower Jurassic Series, Hettangian Stage. The same applies to geochronological units: for example, Jurassic Period, Early Jurassic Epoch and Hettangian Age.

Geochronology and geochronological units

The word geochronology is used in two slightly different ways: first, to denote the science of obtaining numerical, especially radio-isotopic ages from rocks (also termed geochronometry or radio-isotopic dating: Chapter 14); and, secondly, to describe the discipline of subdividing geological time and determining geochronological units: for example, periods and epochs (Table 13.1).

13.2 Organizing rocks and time: chronostratigraphy and geochronology

Chronostratigraphical divisions are rock-time units. They are tangible physical entities that can be studied and measured in the field, being bodies of rock that were formed during a specific interval of time. Chronostratigraphical divisions are bounded by stratigraphical surfaces that are isochronous. Chronostratigraphical classification is the organization of these units based on their age or time of origin.

The time divisions during which the chronostratigraphical units formed are known as geochronological units and are *exactly parallel*. The geochronological units are intangible time units but are defined by, and inextricably linked to, chronostratigraphical units and bear the same given names and hierarchy. We therefore say that strata of the Triassic System (chronostratigraphical unit; Table 13.1) were deposited during an interval of time called the Triassic Period (geochronological unit; Table 13.1; Section 13.3). For stratified rocks, the geochronological division is the time interval over which the rocks were deposited. For nonstratiform rocks (Chapter 2), the geochronology might have several stages: for example, a magma crystallized in the Devonian might be metamorphosed during the Permian. Using chronometric methods, the mineral assemblages and textures formed at different times will be assigned to different geochronological units.

The position within a chronostratigraphical unit is expressed by adjectives indicative of level within the strata, such as, lower, middle and upper. Position within a geochronological unit is expressed by the temporal adjectives: early, middle and late. Both sets of qualifiers are capitalized when referring to formal units, but when used more loosely and generally are uncapitalized. So, for example, we can measure and sample Upper Triassic Series sedimentary rocks, but dinosaurs evolved in the Late Triassic. However, because the Rhaetian Stage is not formally subdivided, we can only say that the upper Rhaetian Stage contains evidence of a carbon isotope excursion, if we wanted to be more specific about the stratigraphical level then a zone or chron or specific height/depth within a section would need to be used.

Table 13.1 The hierarchy of chronostratigraphical and geochronological units

Chronostratigraphical	Geochronological	Typical duration	Comments
Eonothem	Eon	More than hundreds of millions of years	Currently three: Archean, Proterozoic and Phanerozoic, with the first two informally termed the Precambrian
Erathem	Era	Hundreds of millions of years	Consisting of several systems. The Phanerozoic unit names reflect major shifts in the history of life delimited by mass extinction levels and are derived from the Greek: Paleozoic, old life; Mesozoic, intermediate life; Cenozoic (originally Kainozoic), new life
System	Period	Tens of millions of years	Unit of major rank: for example, Paleogene, Cretaceous and Silurian
Series	Epoch	Less than tens of millions of years	Subdivisions between the rank of system/period and subseries/subepoch (if present) or stage/age. Currently the names reflect position (e.g. Upper/Late Cretaceous) or geographical location (e.g. Wenlock), but the recommendation for any *new* series/epoch is a geographical feature or place in the vicinity of the stratotype
Subseries	Subepoch	Less than 10 Myr	Subseries/subepoch are subdivisions of series/epochs and comprise one or more stages/ages. Mainly used in the Cenozoic: for example, Lower/Early Miocene (=Aquitanian+Burdigalian), Upper/Late Miocene (=Tortonian+Messinian). Name has typically reflected position (e.g. Lower/Early or Upper/Late), but the recommendation for *new* subseries/subepoch is a geographical feature or place in the vicinity of the stratotype
Stage	Age	*c.* 2–20 Myr	Name is geographical and ends in -ian or -an: for example, Danian, Maastrichtian and Homerian. Defined by GSSP from which everything else of higher rank follows
Substage	Subage	Hundreds of thousands to millions of years	Currently not formally defined (by GSSP): for example, lower/early Albian.
Non-hierarchical unit			
Chronozone	Chron	Tens to hundreds of thousands of years	A body of rock formed during the time span of a designated stratigraphical unit, typically biostratigraphical or magnetostratigraphical (see Section 4.2.8)

A downloadable version of this table is provided in the online resources as Table S13.1.

All online resources for this chapter can be found at https://doi.org/10.6084/m9.figshare.c.6183919

13.3 Chronostratigraphical and geochronological units

The goal of chronostratigraphical classification is to establish a set of chronostratigraphical units that will serve as a standard scale of reference for the dating of all rocks everywhere and reflect the passage of time (Murphy and Salvador 1999; and the updated online version of Murphy and Salvador 2021). Like most classification schemes, there is a hierarchy of units (Table 13.1), but the basic working unit for formal definition is the 'Stage' (and equivalent geochronological unit 'Age'). This unit was chosen because it is suited in scope and rank to most practical needs and purposes of inter-regional chronostratigraphical classification (Murphy and Salvador 1999, 2021). The duration of stages/ages is variable across the timescale and lies between about 2 and 20 Myr.

13.4 Building a chronostratigraphical scale

Chronostratigraphical names

Chronostratigraphical units are generally named after the locality or region where the rocks of that interval are well preserved. The formal definitions for many boundaries (GSSPs) were defined much later or are still being defined but the historical chronostratigraphical terms are almost always preserved to avoid confusion.

The current approach to the formal development of the chronostratigraphical scale is to identify and describe a primary reference section for the lower boundary of the working chronostratigraphical unit (i.e. a stage), with the upper boundary being defined by the lower boundary of the succeeding unit (Fig. 13.1). This method of defining successive stages forms a continuum of chronostratigraphical units, avoiding gaps and overlaps in the scale (Remane 1997, 2003; Smith *et al.* 2014). The stage boundaries serve to define all chronostratigraphical units of higher ranks (series, systems and erathems), which, when all defined and ratified, will provide a global chronological framework into which rocks and any hiatuses can be unambiguously located.

These reference sections for the stages are boundary stratotypes and combined with a physical reference level in the chosen section are known as GSSPs (Fig. 13.1; Section 13.5). The level in the boundary stratotype section also defines the extent of the geochronological unit (age). The chronostratigraphical unit boundaries are synchronous horizons, by definition, but, in practice, their identification is subject to the same uncertainty that applies to all stratigraphical correlation techniques. For this reason, all possible methods are applied to defining the bounding level of the GSSP, to provide the optimum chance for global correlation. The boundary level is always defined in the GSSP by a carefully chosen *primary* stratigraphical guide or marker (e.g. the base of a fossil distribution (range) or a magnetozone boundary); in addition, it must also fall within an interval of multiple *secondary* stratigraphical markers, which all add to the potential for correlatability (Fig. 13.1; sections 13.5.3 and 13.5.4).

The choice and definition of GSSPs requires worldwide study of the stratigraphical interval, above and below the boundary level, by a scientific working group (Section 13.5.2). The best section and level is chosen by consensus, but must meet a set of well-defined criteria, which include sections that are accessible, stratigraphically complete, non-tectonized and fossiliferous (Section 13.5.4). The *point* should lie within an interval of uniform deposition defined and bracketed by the most comprehensive set of stratigraphical guides possible, and traditionally, but not always, including fossils (Fig. 13.1; Section 13.5.4; Worked example 13.1). It is important to note that while the choice of most GSSPs has happened relatively recently or is still ongoing, this process builds on decades, if not centuries, of stratigraphical study and incorporates knowledge of stratigraphical (and palaeontological) successions not just in and around stage boundaries, but throughout geological records. Sections and points are chosen as boundary standards but critical context is provided by a comprehensive understanding of a diverse range of stratigraphical guides throughout the rock record that enable correlation. Biostratigraphy has been and continues to be one of the most powerful tools for correlation regionally and globally, but advances in chemostratigraphy, cyclostratigraphy and magnetostratigraphy are leading to a rebalancing of emphasis and choice of primary marker.

The International Commission on Stratigraphy (ICS), through its various subcommissions and working groups, is responsible for the international collaboration necessary to achieve global agreement on the choice of individual GSSPs (Section 13.5.1).

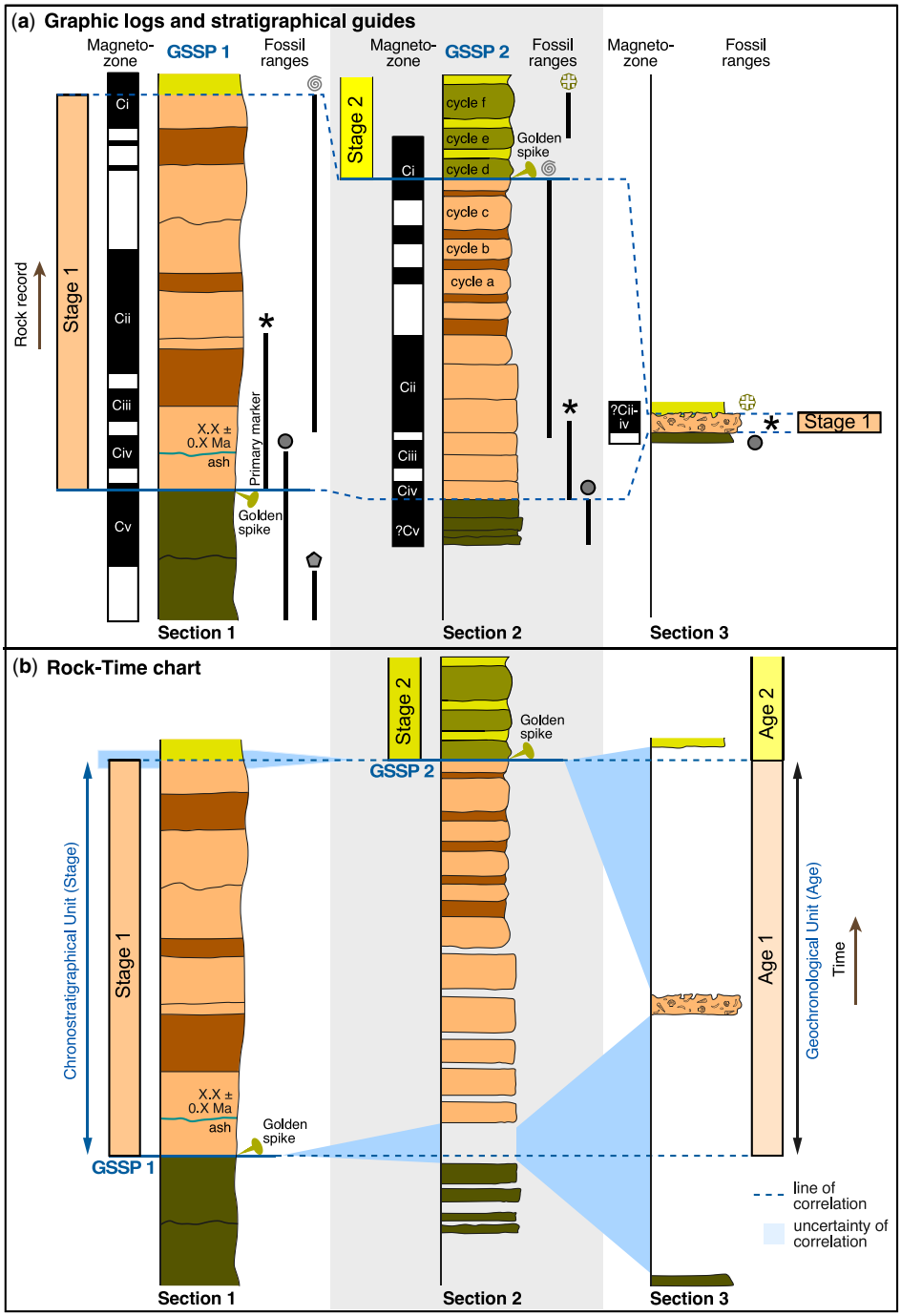


Figure 13.1 Three geological successions shown as (**a**) graphic logs in depth/height with stratigraphical data and (**b**) as a rock-time, chronostratigraphical, diagram. Correlation between these sections relies on a chronostratigraphical framework throughout the sections. Section 1 is a near-continuous succession with a GSSP ('golden spike': see Section 3.6) at its base; Section 2 is a succession with the succeeding GSSP at its top, but hiatuses in the lower part. The bases of Stage 1 and Stage 2 are precisely located at their type sections (GSSPs 1 and 2) and together they define the extent of Stage 1 and duration of the corresponding Age 1 (see b). There is uncertainty when correlating away from the respective type sections, shown by the blue lines becoming dashed. Note also the hiatuses shown by gaps in the lithology column of (b). The pale-blue shading shows all of the uncertainty. Section 3 is a condensed deposit that is not well constrained stratigraphically, but it can be assigned to Stage 1 based on its fossil content (and thus was deposited during Age 1). In this hypothetical example, correlation between the sections is best achieved using biostratigraphy, but the ability to identify magnetozones, astronomical cycles (cycles a–f) or use other criteria such as sequence stratigraphical cycles and chemostratigraphy (not shown) brings additional correlation power and precise age calibrations, as does a radio-isotopically dated ash layer.

Worked example 13.1: Constructing a chronostratigraphy – the Sandbian Stage, Upper Ordovician

This worked example demonstrates four aspects of GSSPs: first, that they are based on a large volume of chronostratigraphical data built up globally; secondly, that an in-depth understanding of the overall succession is critical before a primary marker can be selected; thirdly, that even with the selection of a good primary section, auxiliary stratotype sections are often necessary to aid global correlation (Section 13.5.4); and, finally, the ranges of taxa and their identifications may change but the position of the GSSP is fixed.

The Sandbian GSSP and, hence, the base of the Upper Ordovician is at the Fågelsång section in Sweden. The ranges of fossils, principally graptolites, but also chitinozoans and conodonts, were intensively collected through the Dicellograptus Shale in southern Sweden (Bergström *et al.* 2000). The ranges of these fossils allowed correlation of and within the Dicellograptus Shale in Sweden, but also provided a context to define a GSSP for the base of the Middle Ordovician and a new global stage, the Sandbian (Fig. 13.2). The easily recognizable, near-globally distributed and well-known graptolite *Nemagraptus gracilis* appears within the formation and marked a level where the Sandbian GSSP is defined. Close to the boundary is a radiation of early dicellograptid species. The boundary is marginally above the transition between the *Pygodus serra* and *Pygodus anserinus* conodont biozones. However, the disadvantage of this well-exposed and accessible section is that the deep-water facies makes correlation into many platform successions difficult. To overcome this, auxiliary stratotype sections, in different facies, were nominated in Dawangou (Xinjiang, China) and Calera (Alabama, USA) (Bergström *et al.* 2000).

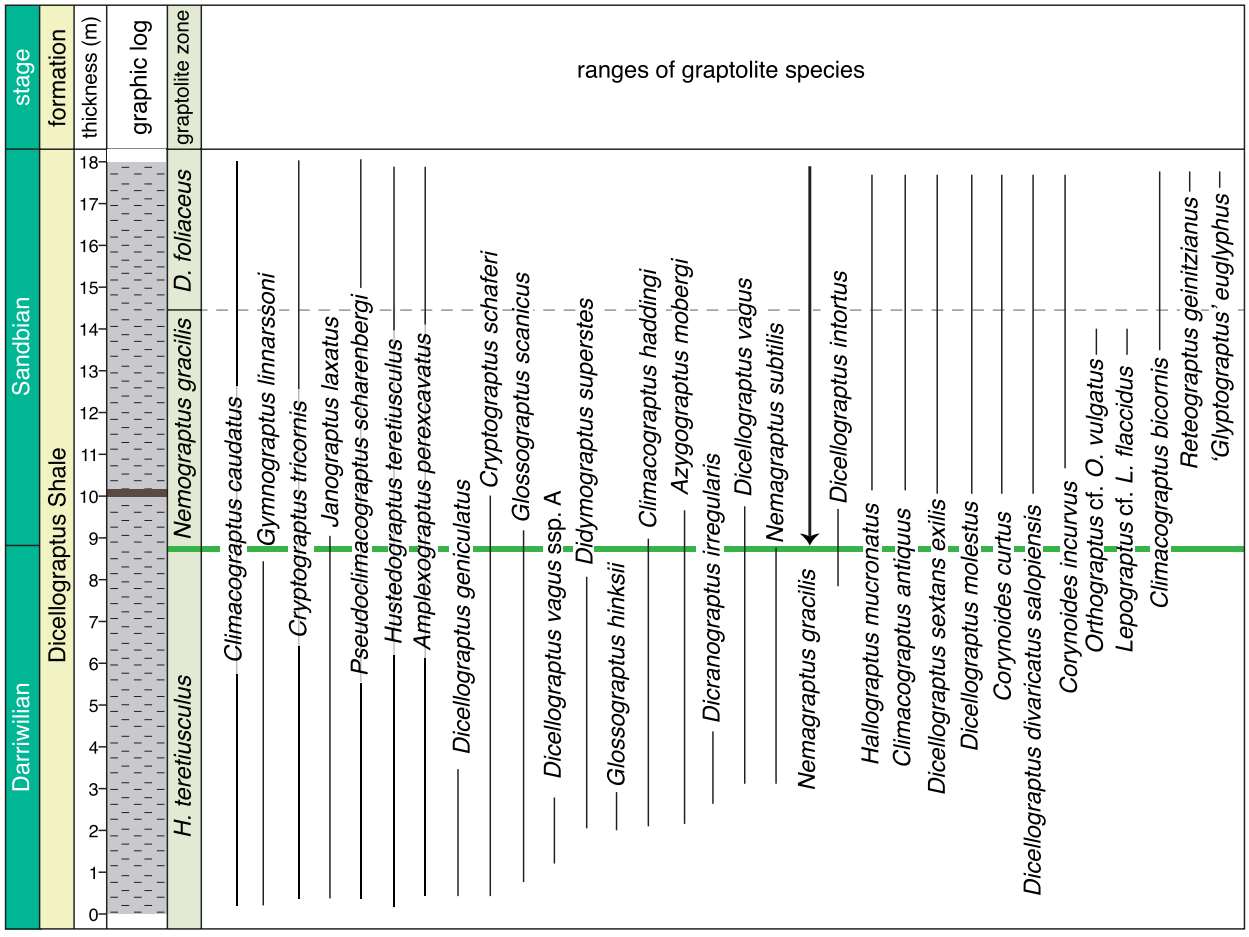


Figure 13.2 Stratigraphical ranges within the *Hustedograptus teretiusculus*, *Nemagraptus gracilis* and lowermost *Diplograptus foliaceus* graptolite biozones, Fågelsång area, Sweden, showing the biostratigraphical context of the Sandbian GSSP. The base occurrence of *Nemagraptus gracilis* is the primary marker for the GSSP (marked as a green horizontal line) and the base of the *Nemagraptus gracilis* graptolite Zone. This point is about 1.4 m below the Fågelsång phosphorite (black bed just above the 10 m level). If the range data were to change, the GSSP remains in place, defined in rock, but the means of correlation requires adjustment. Replotted from Bergström *et al.* (2000, fig. 4).

13.5 Global Stratotype Sections and Points (GSSPs)

Table 13.2 Key points for Global Stratotype Sections and Points (GSSPs)

The GSSP is the internationally agreed reference place and stratigraphical level of the stage/age
The GSSP defines the base/start of the stage/age
The base of erathem/era, system/period, series/epoch always coincide exactly with the base of a stage/age
The top of a stage/age is defined by the base of the overlying stage/age
GSSPs must have primary markers (often biological) and secondary markers
Auxiliary reference sections may also be formally recognized

Many of the units of the International Chronostratigraphic Chart (Section 13.1.1) were originally defined during the nineteenth century at major biological and geological events; these units were commonly associated with hiatuses and consequently have been shown to be incomplete. Hence, modern work focuses on, first, identifying levels and places where the succession representing the boundary interval is the most complete. Secondly, rigorously defining the base of each chronostratigraphical unit (and the beginning of the corresponding geochronological unit) as a GSSP (Table 13.2; Section 13.1) or reference level. Importantly, as outlined in sections 13.3 and 13.4, the top (end) of a unit is automatically defined by the base (beginning) of the suprajacent unit and the stage is the basic working unit from which all the other units in the hierarchy are defined. The GSSP reference levels are colloquially termed 'golden spikes' (Fig. 13.1), borrowing from the pioneers of the North American railway system where a golden spike was driven into the ground to join up the western and eastern lines of the first transcontinental railway across the USA in 1869.

The GSSP marks the unique place and stratigraphical level where a specified chronostratigraphical boundary is defined. Sections elsewhere are then correlated with this, using all available methods. Hence, the GSSP must be at a level and locality that has maximum possible applicability for global correlation. In practice, however, a single data type from a single section is unlikely to be directly correlated throughout the whole world, so as well as secondary markers based on other data types at the GSSP (Section 13.1), supplementary or auxiliary reference locations in different facies are often necessary to provide the global reach (Section 13.5.4). The relative position of primary and secondary markers in the stratotype can also be used to test the reliability of the primary marker in other sections.

13.5.1 The role of the International Commission on Stratigraphy

The ICS (https://stratigraphy.org/) is a permanent commission of the International Union of Geological Sciences (IUGS: https://www.iugs.org/). Its primary objective is to define precisely a hierarchical set of global chronostratigraphical units (Systems, Series and Stages). These units, in turn, define the age-equivalent periods, epochs and ages of the geological timescale. Setting an international global standard is fundamental for expressing geological knowledge and provides a common, transnational language for geological time. It is also of considerable pragmatic importance as it provides the framework through which regional-scale higher-resolution divisions can be linked, equated and collated.

The First International Geological Congress (IGC) in Paris, 1878, initiated various bodies to develop international cooperation in stratigraphy, its classification and terminology. The 11th IGC in Stockholm established a Lexicon of Stratigraphy. However, the Commission on Stratigraphy was formally named at the 19th IGC in Algiers in 1952, with two subcommissions (Lexicon and Terminology subcommissions; both now comprise the Subcommission on Stratigraphic Classification) and the Commission has continuously functioned since, with the addition of more subcommissions, notably one for each system. The Commission applied for admission to the IUGS in 1965; membership was restricted to its officers and presidents (now chairs) of the subcommissions, and this membership continues to present day. The Commission currently comprises 15 system-specific subcommissions, together with one devoted to stratigraphical classification (see above) and one focused on the time calibration of the International Chronostratigraphic Chart. The statutes for the constituent bodies and the rules for formally defining a GSSP are outlined on their webpage (https://stratigraphy.org) and references therein.

13.5.2 Forming a working group and establishing the context of a GSSP

For each geological system, the relevant subcommission forms working groups for each GSSP that needs defining, usually comprising around 20–30 people with a wide range of expertise in different stratigraphical methods. Collectively, the group should have knowledge of relevant sections worldwide, considerable research experience and

Datasets

Investigations for GSSPs yield large multidisciplinary datasets. Consider how this is best organized by multiple authors for multiple users both during the work to define the GSSP and after. It is likely to involve both databases and a suite of linked publications. Chapter 16 provides guidance on curating and archiving a diversity of stratigraphical data in a way that makes it accessible and open to all.

understanding of the wider interval in which the boundary interval sits. The group will work together to formulate and propose one or more possible stratotype section(s) (Fig. 13.3). In their global search for stratotypes, an initial key phase is the investigation of all possible sections, their stratigraphical completeness, and their potential for correlation by both biological and non-biological means (Remane *et al.* 1996) (Fig. 13.3). During this exploratory phase, research data of significant scientific value are gathered. For example, the search for adequate sections to define the base of the Cambrian System has furnished much critical information about the diversification of animal life in this critical interval of macroevolution (Landing *et al.* 2013); and the Ordovician–Silurian boundary generated a large global dataset of time-series information, both biological and non-biological (including isotopic and sedimentological data), across over 50 boundary sections (Cocks and Rickards 1988), critical to understanding the causes and consequences of the Late Ordovician Mass Extinction Event (Harper *et al.* 2014).

13.5.3 Identifying a means of primary correlation

Currently chronostratigraphical units are correlated primarily, but not exclusively, by the range of a fossil organism, a boundary being placed at its base or first appearance (see Section 4.2.8; Fig. 13.1). The stratigraphical ranges of fossils, or assemblages of fossils, form the basis for the definition of biozones, the main operational units of biostratigraphy (Chapter 4). However, the use of such zone fossils is not without problems (Section 4.2.8). For example, there can be difficulties associated with the preservation or identification of some organisms used as zone fossils, and there are often challenges associated with determining the global range of a fossil or a fossil assemblage, or whether the index fossil is found at its lowest/oldest level (Section 4.2.8, Fig. 4.9; e.g. see the case in Hodges 1994 for what was at the time a candidate GSSP). Nevertheless, in the vast majority of cases, biostratigraphy is still usually the preferred means of correlating and establishing the relative ages of strata because of ease of use and the high resolution it can provide. Non-biological markers are becoming increasingly recognized as good primary markers, especially isotopic events, palaeomagnetic reversals, astronomical cycles (Worked examples 13.2) and, in one instance, the deposits from a meteorite (Danian GSSP: Molina *et al.* 2006).

13.5.4 Nominating a section and point for a GSSP

The boundary level is almost always defined in an appropriate rock succession at a particular location (*section*) at a particular stratigraphical datum or level in that one section (*point*). The boundary stratotype sections must pass important criteria set by the ICS (Salvador 1994; Remane *et al.* 1996; Murphy and Salvador 1999, 2021) to be adopted as a GSSP and ratified by the IUGS. Here we distil these criteria into 12 key points arranged in logical but not priority order:

Twelve key points for GSSP

A downloadable version of these twelve key points for a GSSP is provided in the online resources as Table S13.2.

(1) **Lower boundary and working unit**: The GSSP defines the lower boundary of a chronostratigraphical unit. The stage is used as the standard working unit, and the series and systems follow at exactly the same stratigraphical level.

(2) **Well studied**: All possible sections should be well studied. This process usually involves scientific meetings and field visits, alongside publications and shared compilations of data.

(3) **Historical context and precedence**: Historical priority, usage and stratigraphical placement of the boundary should all be taken into account when choosing and justifying the *section* and *point* for the GSSP (but see also the other points especially point 6).

(4) **Correlation potential**: The primary overriding criterion for an excellent GSSP is that the *section* and *point* chosen presents the greatest potential for global correlation now and in the foreseeable future. The potential for correlation can be judged by criteria 5–12, which relate to identification of the *point* in the *section* by primary and secondary markers, together with the suitability of the exposure.

(5) **Depositional environment and palaeogeography**: Open-marine sections almost always offer the best records because they are more likely to be stratigraphically complete and contain fossils that are distributed over a wide area, thereby

improving correlatability. The palaeogeography should be established, and, ideally, a place and interval should be chosen where and when there were open-marine connections to other faunal provinces, and ocean circulation is sufficient that it will not adversely influence the chemostratigraphy.

(6) **Sedimentology, stratigraphical completeness and sedimentation rate**: The *section* and *point* must be chosen where sedimentation appears to be as complete as possible from the options available. Salvador (1994) stated 'essentially continuous deposition'. Stratigraphical completeness is necessary because hiatuses and condensed sections present complications for correlation as they change in extent laterally, important information is likely to be absent and the succession of events may not be separable. It is noted that there may be some conflict between this criterion and historical precedence (see point 3), but stratigraphical completeness is more important. The rate of sedimentation should be sufficient that successive events can be easily separated. The *point* should be chosen where there is no marked change in facies indicating a change in environmental conditions that could have affected the fossils and chemostratigraphy or even indicate a hiatus. It is advisable to undertake and present a sequence stratigraphical analysis as this can provide useful information on potential hiatuses, condensed sections and the likely distribution of sedimentary facies. The optimal GSSP position is in the lowermost highstand systems tract as this is most likely to be preserved over the widest area (Chapter 8). It should be noted that whilst maximum flooding surfaces may preserve an abundance of widely distributed fossils they also represent a period of condensation (Chapter 8).

(7) **Adequate exposure**: The *section* must represent an adequate period of time above and below the chosen *point* to allow the boundary to be put into stratigraphical context, and for sufficient secondary markers to be identified. There should also be enough lateral exposure for context and to allow for future study and sampling.

(8) **Alteration**: The *section* should not be seriously affected by tectonic (e.g. faulting), synsedimentary (e.g. slumping) or surficial disturbance (e.g. sliding) because these may adversely affect the stratigraphical continuity of the succession. The *section* should not be significantly affected by metamorphism or diagenesis to the degree that the magnetostratigraphical and primary chemostratigraphical signals are destroyed (e.g. dolomitization).

(9) **Fossils**: The fossils in the *section* should be abundant, distinctive, well preserved, and as cosmopolitan and diverse as possible. This is because fossils provide one of the best means of correlating beds and determining their relative age.

(10) **Primary marker**: A primary marker must be chosen to mark the *point* of the GSSP. This marker should be suitable for at least regional and, if possible, global correlation. Usually, the primary marker is the first/lowest appearance of a fossil species. However, non-biological markers can also be used (e.g. Ediacaran Stage and Meghalayan Stage: Worked example 13.2). The marker should also be independent of facies (see point 5).

(11) **Secondary markers**: There should be as many secondary markers as possible, close to the chosen *point*. These can, and usually do, include several other bioevents across a range of fossil groups. There must also be as many non-biological secondary markers as possible, usually all or most of: distinct chemical signatures such as a changes in carbon or strontium isotopes ratios (note point 5), one or more magnetochron boundaries, astronomical cycle boundaries and ash layers, minerals or deposits that have been radio-isotopically dated (Chapter 14). Where there is more than one *section* to choose from, the number of secondary markers should be considered.

(12) **Accessibility and protection**:

- The *section* has to be freely accessible for all researchers and other interested parties of all nationalities.
- The *section* has to be located where it can be visited efficiently without the requirement for undue resources.
- The *section* must be maintained in good condition (ideally a national reserve) in an accessible terrain.

Hiatuses

When proposing a GSSP consider the nature of the event and how instantaneous it might be, as well as the likelihood of hiatuses or diachroneity at the scale of interest (chapters 8, 9 and 12).

- Fossils and other relevant samples should be archived, stored securely in a permanent facility and accessible for study.
- The GSSP should be indicated by a permanently fixed marker (e.g. Fig. 13.4).

We recommend here that cyclostratigraphical analysis and, where possible, orbital tuning is considered due to the rapid progress in this method and the extra insights it can provide on the duration between the different markers and the temporal extent of the *section* (chapters 9 and 15). We also note that some pragmatism may be required as it is unlikely that there is ever a perfect candidate, as demonstrated by many GSSPs ratified to date.

Auxiliary stratotype sections: these can considerably enhance the value and support the GSSP and have been used in cases where not all the criteria can be met in one section or a further reference section is needed to improve correlatability: for instance, in a different facies (Worked example 13.1), climatic zones (Worked example 13.2) or in a different palaeogeographical context. These auxiliary sections covering the boundary interval should be correlated to the GSSP by more than one marker. Auxiliary sections are subordinate to the GSSP. Head *et al.* (2022) recently presented a good case and recommend that the hitherto formally recognized Auxiliary Stratotype Point is discontinued because whereas further sections can be beneficial there can only ever be one point as defined by the GSSP. A proposal to instead only formally recognize auxiliary sections (Standard Auxiliary Boundary Stratotype) is being voted on by the ICS at the time of writing.

One or more GSSP proposals should be prepared. Following discussion and voting (yes/no/abstain) on one or more proposals within the working group, the proposal(s) will then need to pass through several further discussion and voting phases (Section 13.5.5; Fig. 13.3).

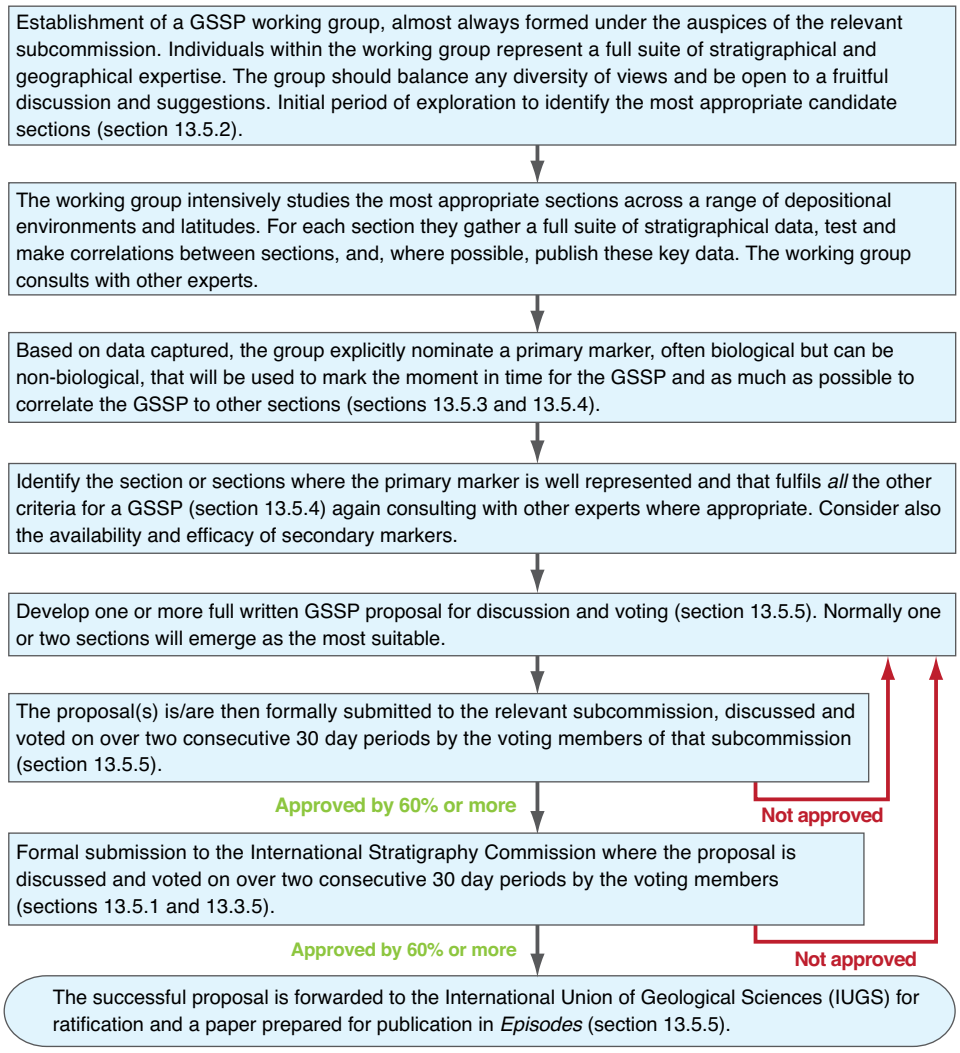


Figure 13.3 Flow chart for the process of defining a GSSP.

13.5.5 The route to GSSP approval

To finalize the choice of a GSSP, a detailed proposal must be discussed and presented that outlines its global utility and discusses its compliance with the 12 points outlined in Section 13.5.4. This should include a brief summary of the sections and points considered. During the journey of proposals to ratification, various groups within the ICS act as critical friends as summarized in the flow chart shown in Figure 13.3. Following voting by the subcommission's working group (Section 13.5.2) the group will present its proposal and results to the relevant subcommission. The process of discussion and voting will be repeated here, and when complete the same procedure will be followed in the ICS. On reaching a supermajority of 60%, at all stages in the process, the final proposal is escalated to the Executive Council of the IUGS for ratification and, if successful, the GSSP will be added to the International Chronostratigraphic Chart. The accepted proposal is published in the journal *Episodes* (Fig. 13.3).

GSSP proposals

To get an idea about how to structure and write a good proposal, consult a range of accepted proposals published in the journal *Episodes*. You can use the search term GSSP in *Episodes* to find these publications.

The *point* in time, indicated by the GSSP, is a fact. It marks an instant in geochronological time. All else, including its correlation, is interpretation. Worked example 13.2 briefly illustrates some of the key elements of the definition and correlation of GSSPs and how solutions have been found to a variety of challenges.

13.6 The Archean and Proterozoic timescale and Global Standard Stratigraphic Ages (GSSAs)

The standard GSSP approach (Section 13.5) is not yet widely applied in the Archean and Proterozoic part of the timescale, largely because relative correlation is difficult in the absence of fossils. Instead, most boundaries (except for the Ediacaran System: Worked example 13.2) are currently defined in terms of numbers of years, using geochronometric units called Global Standard Stratigraphic Ages (GSSAs). For example, the Hadean–Archean boundary is set at 4 Ga and the Archean–Proterozoic boundary at 2.5 Ga. These GSSAs will likely be mostly superseded by GSSPs, with better knowledge of the successions and once suitable GSSP candidate levels are recognized. Such levels are likely to be chosen using geochemical proxies that reflect important palaeoenvironmental changes (Shields *et al.* 2021). For instance, the Archean–Proterozoic boundary roughly coincides with the oxygenation of Earth's atmosphere, and so potentially a geochemical signal reflecting that change might serve as candidate, if it can be shown to be globally correlatable.

Worked example 13.2: Solutions used to deal with challenges associated with defining some GSSPs

In the worked examples below, five different solutions to defining and correlating GSSPs are shown, within the criteria outlined in Section 13.5.4 and references therein.

Highlighting a solution where there were no biological markers: GSSP for the Ediacaran System

The Ediacaran is the highest of the Proterozoic systems and the longest of the GSSP-defined geological periods (635–541 Ma). Defining and correlating the base of the system was challenging, given the lack of biostratigraphical markers and limited non-biological data. The Ediacaran GSSP exemplifies the possibilities of defining and correlating GSSPs in deep time in the absence of biological markers, and opens up the possibility of replacing GSSAs with rock-based definitions of more Archean and Proterozoic chronostratigraphical units.

Context The Ediacaran is a critical time in Earth's history, beginning with the rapid decay of the Marinoan ice sheets; it is characterized by a distinctive pattern of secular changes in carbon isotopes and in its upper part by the macro-scale, soft-bodied Ediacaran biota (Knoll *et al.* 2006). Although historically the Ediacaran was largely based on the occurrence of this biota, these organisms appear in force relatively late during the period, after the Gaskiers Glaciation (at *c.* 580 Ma). Investigation of the Ediacaran over many decades, principally in search of its fossils, has accumulated significant quantities of stratigraphical data together with biological and non-biological time-series data.



Figure 13.4 The fixed marker (a metal disk) for the Ediacaran GSSP at the base of the Nuccaleena Formation 'cap carbonate', southern Australia. The exact level bisects the middle of the disk (see Xiao *et al.* 2016). Image is about 1 m high. Image from Shuhai Xiao.

Definition and correlation There are no biological markers that are currently suitable as a primary means of correlation for the GSSP. The appearance of organic-walled microfossils, notably acanthomorph acritarchs, has yet to be fully analysed for boundary correlation, whilst the Ediacaran soft-bodied biota occurs too high in the system (Xiao *et al.* 2016). Instead, the following two primary markers were used to define the GSSP: (1) the distinctive lithological change from deposits of the Marinoan glaciation (Elatina Formation) to the Nuccaleena Formation cap carbonate (Fig. 13.4); and (2) the characteristic upward decrease in carbon isotope ratios in the Nuccaleena Formation (Knoll *et al.* 2006). Similar cap dolostones are found in many places throughout the world (Hoffman *et al.* 2017), and this lithological change together with the carbon isotopes provide an excellent level for correlation worldwide. The base of the system has been dated elsewhere in China and Namibia at 635 Ma (see Xiao *et al.* 2016).

Location The GSSP is at the Enorama Creek section, Flinders Ranges, South Australia (Fig. 13.4) at the base of the 6 m-thick Nuccaleena Formation cap carbonate.

Highlighting a solution that combines biostratigraphy and chemostratigraphy for global correlation: GSSP for the Hirnantian Stage

Positioning the base of the Hirnantian was based on study of graptolites and the *Hirnantia* brachiopod fauna in over 40 sections in China where this interval is well represented, in addition to sections in Europe and elsewhere in Asia, Africa together with North and South America. Global correlation is also facilitated by the distinctive, global carbon isotope excursion.

Context The Hirnantian is the highest stage of the Ordovician System. The term Hirnantian was first used as a stage name by Bancroft (1933) for the highest Ordovician strata in the Bala district, North Wales. He considered that the Hirnantian was associated with the widespread *Hirnantia* brachiopod fauna including '*Orthis*' *sagittifera* [*Hirnantia*] and '*Rafinesquina*' *hirnantensis* [*Eostropheodonta*]. The lower part of the Hirnantian marks the first of the big five Phanerozoic mass extinction events, in this case associated with the waxing and waning of a major polar ice cap. The focus on the Hirnantian and the overlying Rhuddanian Stage has provided a huge amount of biological and non-biological data on the terminal Ordovician extinction, its causes and consequences (e.g. Rong *et al.* 2020).

Definition and correlation At the point chosen, the base of the Hirnantian Stage is coincident with the first appearance of the graptolite *Metabolograptus extraordinarius* (Sobolevskaya) and is close to the first appearance of the widespread *Hirnantia* brachiopod fauna. The entire Hirnantian stratigraphical succession is well developed in the Wangjiawan North section (Hubei Province, China), as is the base of the Silurian, which defines the top of the Hirnantian Stage. In addition to the graptolitic and shelly faunas, the boundary lies near the base of the positive Hirnantian Isotope Carbon Excursion (or the HICE); this chemostratigraphical signature has proved critical for correlating the boundary. The GSSP section can be correlated globally in all geological settings using either biological signals (graptolites and brachiopods) or carbon isotopes.

Location The GSSP was ratified in 2003 and defined at a stratigraphical level 0.39 m below the base of the Kuanyinchiao Bed in the Wangjiawan North section (Chen *et al.* 2006).

Highlighting the importance of secondary markers due to provincialism of primary biological markers: GSSP for the base of the Hauterivian Stage

The Hauterivian is the third stage in the Lower Cretaceous Series. Unusually, the GSSP is placed at the location where the stage was first defined (Mutterlose *et al.* 2021). Correlation of the GSSP is based primarily on ammonites, although the ammonite correlation is far from straightforward due to provincialism: that is, the species are not present outside of the biogeographical Tethyan Realm of southern Europe and North Africa. This section and point was accepted

as the best that could be found, with secondary markers providing the potential for global correlation (Fig. 13.5).

Context The Hauterivian Stage was originally recognized in the area of Hauterive (Neuchâtel, Switzerland), and includes from the base to the top three lithological units: the Marnes à Astieria (later included in the underlying Valanginian), the Marnes d' Hauterive à Ammonites Radiates and the Pierre Jaune de Neuchâtel. The stage, however, was first defined at La Charce in southern France, where the GSSP is now located. The primary biostratigraphical marker is the first appearance of the ammonite *Acanthodiscus rebouli*. Ammonites are a widely used fossil group in Jurassic and Cretaceous biostratigraphy due to their rapid evolutionary turnover and relatively frequent occurrence.

Definition and correlation At the GSSP and in the Tethyan biogeographical realm, the base of the Hauterivian is recognized by the first appearance of the ammonite genus *Acanthodiscus*, here represented by the species *A. rebouli*, and base of the *Acanthodiscus radiatus* ammonite Zone. These ammonites are not present in other biogeographical realms, and in the Boreal Realm the base of the stage is approximated using a migration of ammonites into the province where the lowermost part of the boreal *Endemoceras amblygonium* ammonite Zone corresponds to the uppermost part of the *Criosarasinella furcillata* ammonite Zone of the La Charce section. The Weissert Carbon Isotope Excursion reaches its maximum below the boundary and $^{87}Sr/^{86}Sr$ values increase from 0.707355 to 0.707379 near the base of the Hauterivian. The age of the base of the stage is 132.6 Ma based on cyclostratigraphy and spline fitting.

Location The GSSP for the Hauterivian Stage is defined in the commune of La Charce, Drôme, southern France at the base of Bed 189 where it occurs within moderately deep marine facies, deposited in the Vocontian Basin (Fig. 13.5).

Highlighting the advantages of multiple stratigraphical markers: GSSP for the base of the Chattian Stage

The Chattian is the uppermost stage in the Oligocene Series, overlying the Rupelian and underlying the Aquitanian of the Miocene (Figs 13.6 and 13.7).

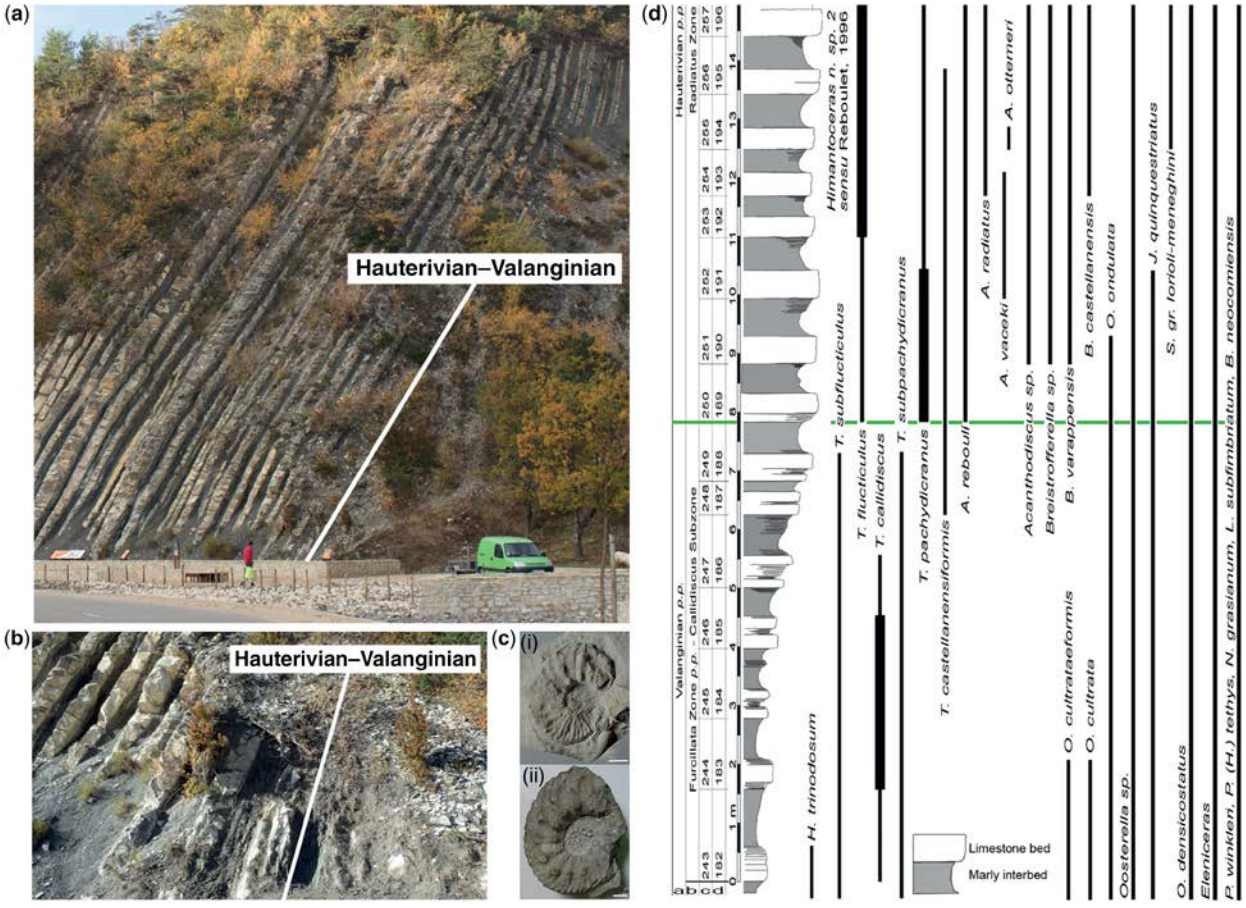


Figure 13.5 The Hauterivian GSSP at the base of Bed 189 in the commune of La Charce, Drôme, southern France: **(a)** general photograph of the section; **(b)** detail of the lower part of section just above the road; **(c)** (i) *Acanthodiscus rebouli*, the GSSP biostratigraphical marker; and (ii) *Acanthodiscus radiatus* (zonal species, Bed 198): the scale bar is 1 cm; and **(d)** graphic log and ammonite distribution with the GSSP marked by the green line. Compiled from parts of Mutterlose *et al.* (2021, figs 2, 3 and 4).

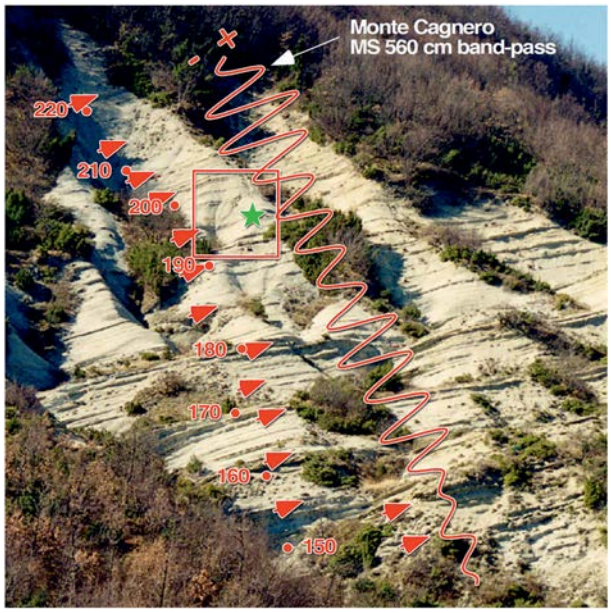


Figure 13.6 The Chattian GSSP at 197 m above the base of the Scaglia Cinera Formation, Monte Cagnero, Italy (indicated by an asterisk). Also shown are the metre levels (spots and numbers) and the 5.6 m band-pass magnetic susceptibility minima (arrows) and curve. See Figure 13.7. Modified from Coccioni *et al.* (2018, fig. 4a).

The definition of this GSSP was determined only after multiple biological and non-biological records were assembled, during many years of research, and integrated to provide a chronostratigraphy through the stage and the strata below. The historical stratotype section in NW Germany was not proposed as the GSSP because of the marginal-marine facies that did not provide adequate plankton microfossils, palaeomagnetic signals or radiometrically datable material. The working group carried out an extensive search for more suitable sections and focused on three deeper-water localities in Italy (Coccioni *et al.* 2018). The Monte Cagnero section was finally chosen as the rhythmic marls and limestones contain multiple stratigraphical markers, including plankton microfossils (nannofossils, planktonic foraminifera and dinoflagellate cysts), datable volcano-sedimentary horizons, and magnetostratigraphical, cyclostratigraphical and chemostratigraphical (strontium, carbon and oxygen isotopes) records. This enabled the development of a fully integrated age model (Coccioni *et al.* 2018). The GSSP for its base was ratified in 2016 and published by Coccioni *et al.* (2018).

Context The GSSP was chosen in the Monte Cagnero section near Urbania within 86 m of rhythmically bedded marls and marly limestones of the Scaglia

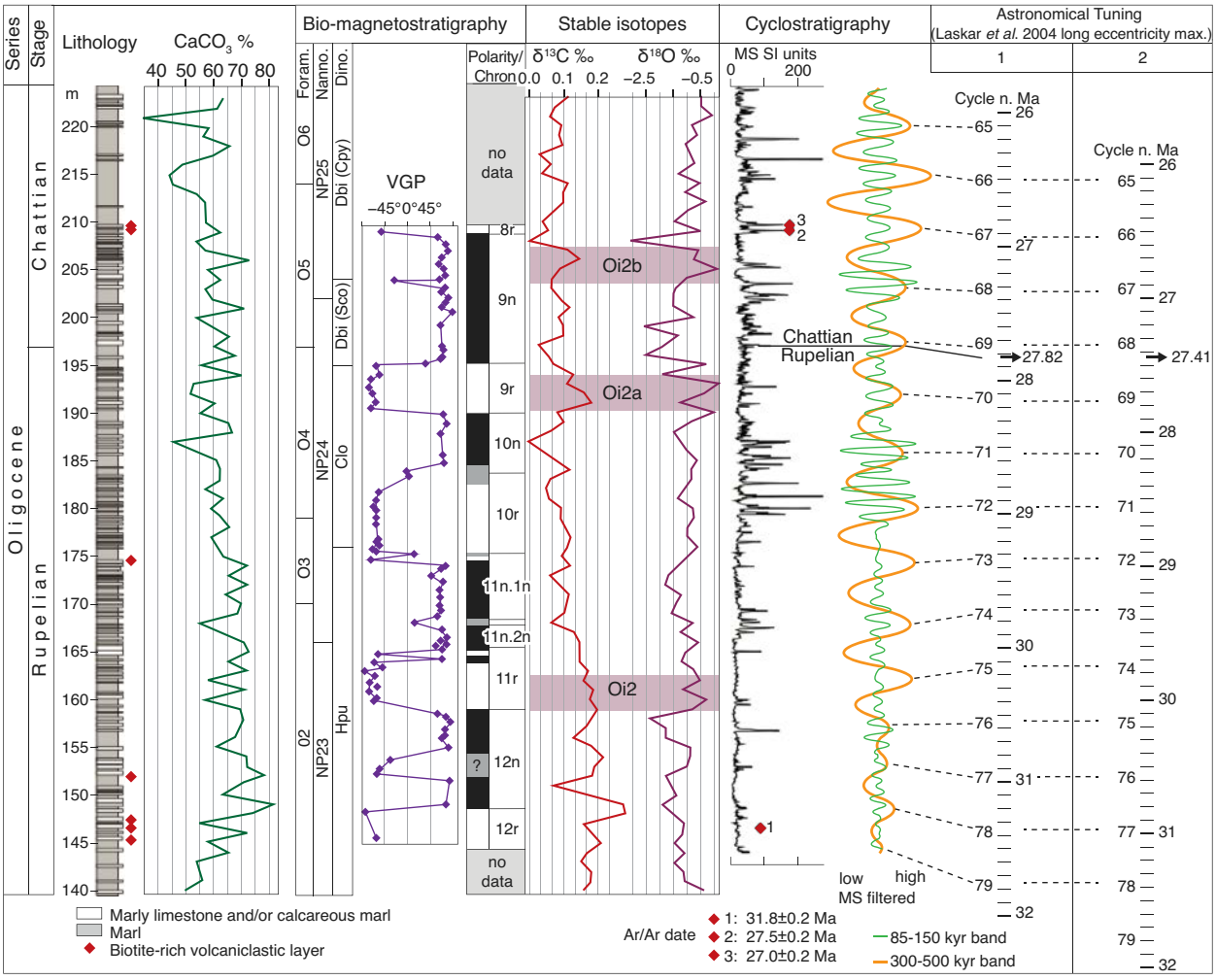


Figure 13.7 The Chattian GSSP is characterized by multiple markers and is orbitally tuned; the chart presents an integrated chonostratigraphy for the Rupelian and Chattian stages. Redrawn from Coccioni *et al.* (2018, fig. 3).

Cinera Formation. The alternating lithologies contain regular cycles and a primary magnetostratigraphical record, while volcaniclastic layers have facilitated radio-isotopic dating.

Definition and correlation The GSSP is placed at the last (or top) common occurrence of the planktonic foraminifer *Chiloguembelina cubensis*, which also defines the base of the O5 planktonic foraminifera Zone; this horizon correlates with the lower part of magnetochron C9n and is close to stable isotope excursions related to the Oi2a glacial event. An astrochronological calibration of the section provides an astronomical age of 27.82 Ma for the boundary (Figs 13.6 and 13.7).

Location The Chattian GSSP is placed at 197 m above the base of the Scaglia Cinera Formation, Monte Cagnero, Urbania, Umbria-Marche, Italy.

Highlighting use of evidence other than a rock exposure: GSSP for the base of the Meghalayan Stage

The Meghalayan is the uppermost stage in the Phanerozoic, comprising the top of the Upper Holocene Series. To provide appropriate ultra-high stratigraphical resolution for this part of Earth's history requires the choice of atypical geological archives; in this case a stalagmite that records a distinctive climate event, identified by chemostratigraphy (Fig. 13.8) (Walker *et al.* 2018).

Context The GSSPs for the lower two stages of the Holocene Series, the Greenlandian and Northgrippian, are in archived ice cores, while the youngest, the Meghalayan, is in a speleothem. Although these archives are not in traditional sedimentary successions, the methodology is comparable. The points in the ice cores and speleothem are correlated by non-biological proxies, such as radiocarbon (Chapter 15), U–Th radio-isotopes and luminescence dating, allowing precise correlation. An auxiliary stratotype without a point is also defined for the base of the Meghalayan at high latitude using an ice core from the Plateau ice field on Mount Logan, Yukon, Canada (Head *et al.* 2022).

Definition and correlation The GSSP is placed at 7.45 mm below the tip of speleothem KM-A (Fig. 13.8). The point is a level within the longer climatic event known as the Holocene 4.2 ka climatic event, which lasted for two–three centuries, and was linked to aridification and significant human cultural and societal changes in low to mid latitudes. The GSSP coincides with a shift to heavier $\delta^{18}O$ values, the higher deuterium excess being associated with the increased transport of moisture from the Pacific tropics (Walker *et al.* 2012).

Location The speleothem KM-A grew in Mawmluh Cave, in the state of Meghalaya, NE India. The cave is one of the longest and deepest in India with the ability to grow and preserve substantial speleothems (Fig. 13.8).

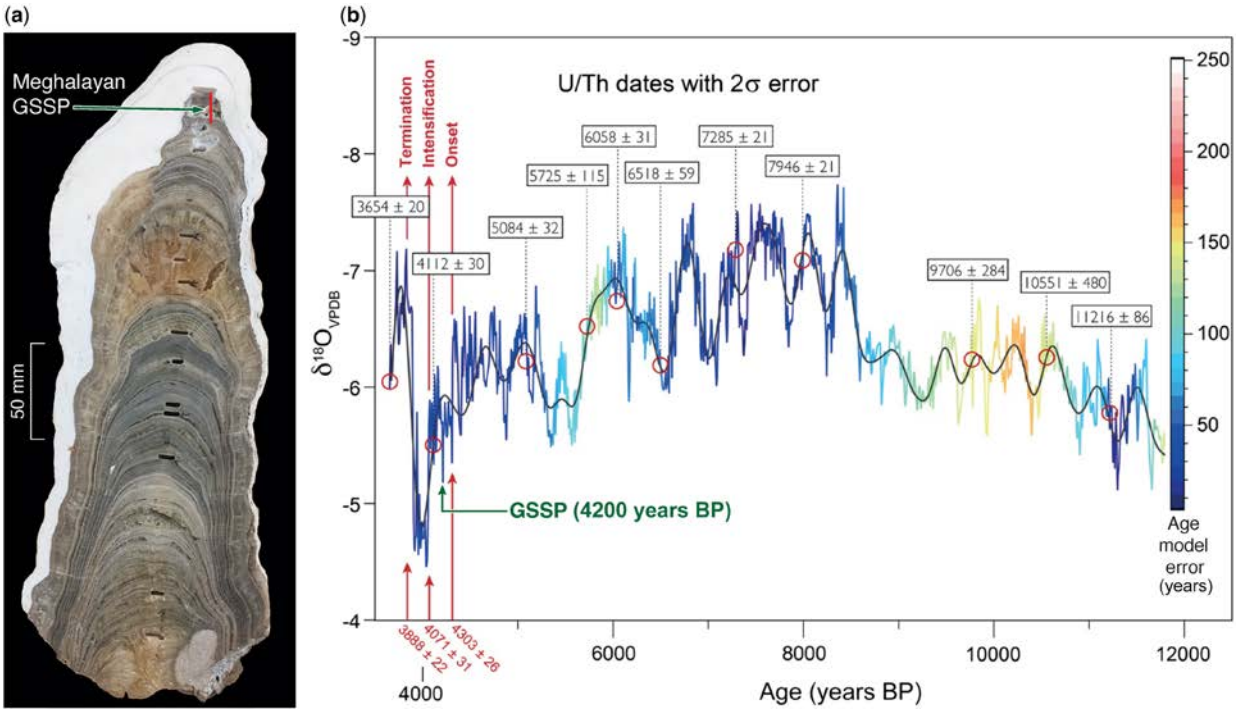


Figure 13.8 **(a)** The Meghalayan GSSP speleothem KM-A from Mawmluh Cave, India and **(b)** associated oxygen isotope and U/Th radio-isotopic data. The GSSP is 7.45 mm below the tip of the speleothem at a shift to heavier $\delta_{18}O$ values (marked by green arrows) and is within an interval that is interpreted to represent a climate event that spanned from 4.31 to 3.65 ka, otherwise known as the 4.2 event (red bar in (a) and red labels in (b)). Modified from Walker *et al.* (2018, figs 5a and 6).

13.7 Regional chronostratigraphical schemes

Regional schemes

Regional chronostratigraphical terms have been used by geologists for over two centuries as 'unit stratotypes'. They appear on maps and on sample labels in museums. But to enhance their global relevance and usefulness, these units should be mapped onto global chronostratigraphies. This is essential, for example, when developing global biodiversity and palaeoclimatology curves together with palaeogeography.

In some parts of the world, localized chronostratigraphical classifications and regional stage names are used for reasons that include: (1) historical precedence of units that were specifically constructed for that region by regional and national geological communities, working together since the eighteenth and nineteenth centuries (e.g. the wide range of names for parts of the Carboniferous: see Gradstein *et al.* 2020, fig. 23.5); (2) and/or insufficient research, to date, on correlating to international standards since work began on these in the 1960s; and (3) extreme difficulty in correlating to global schemes because of basin isolation and/or problematic facies (e.g. Paratethyan Neogene sections). Such regional chronostratigraphical schemes are still used because their boundaries, chosen to best-fit local geological features, are convenient and locally appropriate. However, the resulting plethora of local chronostratigraphical names are generally unfamiliar (and frequently confusing) to geologists elsewhere in the world – and, hence, the necessity for the international standard of the ICS Chronostratigraphic Chart, and a globally applicable geological timescale that provides a common language for all. In practice, a good deal of effort has gone into producing correlation charts between the many regional/national schemes and the global one, and these are invaluable for geologists as they seek to familiarize themselves not just with the geological characteristics, but also the formal classification of new terrains. Many of the regional terms that are regularly used are outlined in Gradstein *et al.* (2020).

13.8 Summary and conclusions

Understanding the relationship between rocks and time (chronostratigraphy) is vital for regional and global correlations, as well as the interpretation of rates, durations and ages from the rock record. This is facilitated by the International Chronostratigraphic Chart and related classification schemes.

The chronostratigraphical and parallel geochronological divisions together form hierarchical frameworks for the International Chronostratigraphic Chart, the basis for the Geological Timescale. The chronostratigraphical hierarchy of eonothem/erathem/system/series/stage/substage is a physical one of 'rock time', most clearly applicable to stratified successions where superposition occurs, and where 'upper', 'middle' and lower' subdivisions can be distinguished. The complementary geochronological hierarchy of eon/era/period/epoch/age/subage of time and inferred events can be applied to all rocks, whether superpositional relationships can be established or not, and can be qualified by early, middle and late subdivisions. The boundaries of both chronostratigraphical and their corresponding geochronological units for the Phanerozoic and some of the Proterozoic are defined by GSSPs. These GSSPs are proposed by international teams of specialists that comprise the subcommissions of the International Commission on Stratigraphy and ratified by the International Commission on Stratigraphy and International Union of Geological Sciences. Whilst traditionally most GSSPs are correlated by biological markers, the application of complementary geochemical, geomagnetic and astrochronological markers is becoming increasingly important and offers solutions that do not have the complications of synchronous global ranges, provincialism and fossil preservation. Most boundaries in the Proterozoic and Archean continue for now to be defined by a numerical date (i.e. GSSAs) but in the future some of those boundaries may be replaced by rock-based definitions (Shields *et al.* 2021).

At the time of writing some 80 Phanerozoic GSSPs are now defined (Cohen *et al.* 2022) and approximately 25 still require attention. Major advances will involve a new set of Proterozoic and Archean chronostratigraphical units, where possible based on GSSPs, and the immensity of Proterozoic and Archean time will be split into shorter more manageable units through the definition of more systems. Throughout the Phanerozoic, radio-isotope ages are being enhanced and refined, and the orbital tuning of sections together with numerical dates offer a real prospect of an accurate chronometrical calibration of chronostratigraphy. We will then be in a much stronger position to understand Earth's history and calibrate its events and processes.

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Chapter 13 image: Zumaia section, northern Spain. The GSSP for the Thanetian Stage of the Paleogene System is near the top of the thick marl layer at the centre of the cliff, at the base of the photograph. Image from Schmitz *et al.* (2011) and photograph courtesy of Juan Ignacio Baceta, The University of the Basque Country.

Online resources for this chapter are available at: https://doi.org/10.6084/m9.figshare.c.6183919